Threshold Output Preservation under Bounded Score Perturbations

Research Architecture Runtime
`feature/lean-gated-operator-papers`

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Abstract

We study the scalar thresholding operator $A_T(x) = \mathbf{1}\{x \geq T\}$ under additive score perturbations with $|x'-x| \leq \varepsilon$. Because equality passes, preservation is asymmetric: $x \geq T+\varepsilon$ guarantees output 1, while $x < T-\varepsilon$ (strict) guarantees output 0. On the half-open unstable band $[T-\varepsilon, T+\varepsilon)$ the decision need not be invariant; we exhibit sharp one-dimensional counterexamples. Publication is gated on Lean `LEAN_FULL` for `Research.Operators.Threshold.Preservation.threshold_preservation` (Mathlib $\mathbb{R}$; `REAL_MATHLIB`).

1 Introduction

Thresholding is a canonical selection primitive in adaptive data analysis and differential privacy (AboveThreshold / Sparse Vector). Before noisy sequential mechanisms, the deterministic scalar rule already exhibits a clear decision boundary $x = T$ and a margin-controlled robustness statement. This note records that statement as an operator-specific theorem exercised end-to-end through Discovery and Verification in this repository.

2 Problem formulation

Fix a threshold $T \in \mathbb{R}$ and a score $x \in \mathbb{R}$. Define

$$A_T(x) = \mathbf{1}\{x \geq T\}.$$

Equality convention. $A_T(x) = 1$ if and only if $x \geq T$ (equality counts as a pass).

Perturbation model. Admissible neighbors satisfy $|x' - x| \leq \varepsilon$ for $\varepsilon \geq 0$. Non-finite scores or thresholds are rejected by the implementation.

3 Notation and definitions

Definition 1 (Signed margin and distance)**.** *$m(x) = x - T$ and $d(x) = |x - T|$.*

Definition 2 (Preservation predicates)**.** *Pass is preserved if every admissible x' satisfies $A_T(x') = 1$. Fail is preserved if every admissible x' satisfies $A_T(x') = 0$.*

4 Mathematical assumptions

1. $x, T \in \mathbb{R}$ are finite.
2. $\varepsilon \geq 0$ and $|x' - x| \leq \varepsilon$.
3. Equality convention $x \geq T$ passes.

We do not treat data-dependent maps $x(D)$ beyond the abstract score, nor Sparse Vector privacy accounting in this note.

5 Main theorem

Theorem 3 (Threshold output preservation)**.** *Under the assumptions above:*

1. *If $x \geq T + \varepsilon$, then $A_T(x') = 1$.*
2. *If $x < T - \varepsilon$, then $A_T(x') = 0$.*
3. *If $x \in [T - \varepsilon, T + \varepsilon)$, the output need not be invariant.*

Theorem 4 (Sharpness with equality asymmetry)**.**

1. *If $x \in [T, T + \varepsilon)$, then $x' = x - \varepsilon$ is admissible and $A_T(x') = 0$.*
2. *If $x \in [T - \varepsilon, T)$, then $x' = x + \varepsilon$ is admissible and $A_T(x') = 1$.*
3. *At $x = T + \varepsilon$, every admissible x' satisfies $A_T(x') = 1$ (pass-side condition is non-strict).*
4. *At $x = T - \varepsilon$, fail is* not *preserved (since $x' = T$ passes).*

6 Supporting lemmas

Lemma 5 (Interval image)**.** *The set of admissible x' is the closed interval $[x - \varepsilon, x + \varepsilon]$.*

Proof. Immediate from $|x' - x| \leq \varepsilon$. □

7 Proofs

Proof of Theorem 3. If $x \geq T + \varepsilon$, then $x - \varepsilon \geq T$, so every $x' \geq T$ and $A_T(x') = 1$. If $x < T - \varepsilon$, then $x + \varepsilon < T$, so every $x' < T$ and $A_T(x') = 0$. If $x \in [T - \varepsilon, T + \varepsilon)$, the interval $[x - \varepsilon, x + \varepsilon]$ intersects both $\{y : y < T\}$ and $\{y : y \geq T\}$ (or, at the left endpoint, reaches T), so both outputs are attainable except on the empty $\varepsilon = 0$ band. □

Proof of Theorem 4. For $x \in [T, T + \varepsilon)$, $x - \varepsilon < T$, hence $A_T(x - \varepsilon) = 0$. For $x \in [T - \varepsilon, T)$, $x + \varepsilon \geq T$, hence $A_T(x + \varepsilon) = 1$. At $x = T + \varepsilon$, $x - \varepsilon = T$ still passes. At $x = T - \varepsilon$, $x + \varepsilon = T$ passes, so fail is not invariant. □

8 Discussion

8.1 Intuition

The threshold is a hard cut. A buffer of size ε on the correct side of T absorbs worst-case movement toward the cut. Because the cut includes equality, the pass buffer is closed on the left ($x \geq T + \varepsilon$) while the fail buffer is open ($x < T - \varepsilon$).

8.2 Geometric explanation

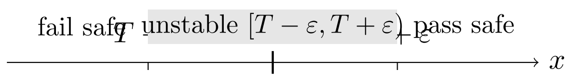

Figure 1: Asymmetric preservation regions for $A_T(x) = \mathbf{1}\{x \geq T\}$.

8.3 Why it matters

A measured margin $m = x - T$ is an operational certificate for selection stability under bounded score noise—the scalar core of AboveThreshold.

8.4 Where assumptions are used

Finiteness excludes NaN/ $\pm\infty$; the equality convention forces the half-open unstable band; $\varepsilon \geq 0$ covers the trivial invariant case $\varepsilon = 0$.

8.5 Examples

Take $T = 0$, $\varepsilon = 1$. Then $x = 2$ and $x = 1$ are pass-safe; $x = 0.5$, $x = 0$, $x = -0.5$, $x = -1$ are unstable; $x = -2$ is fail-safe.

8.6 Counterexamples and boundary cases

At $x = T + \varepsilon$ pass cannot be broken. At $x = T - \varepsilon$ the adversary $x' = T$ flips a fail to a pass.

9 Interpretation

Compare $|m|$ to ε with the stated asymmetry before trusting a reported pass/fail bit under score uncertainty.

10 Utility implications

Abstention bands use the same asymmetry as preservation: release 1 only when $x \geq T + \tau$, release 0 only when $x < T - \tau$, else $\perp$. (The naive closed fail rule $x \leq T - \tau$ is *not* robust at $x = T - \tau$ when $\varepsilon = \tau$.) Set-valued selectors $\{0\}$, $\{1\}$, $\{0, 1\}$ match the preservation predicates.

11 Noisy and sequential extensions

Noisy mechanisms (query noise, threshold noise, both), abstention, and first-crossing $\widetilde{\tau}$ are implemented as a portfolio. A companion note verifies bounded almost-sure noise pathwise. Full Sparse Vector privacy is **not** claimed here.

12 Limitations

- Lean formalization is Mathlib $\mathbb{R}$ (`REAL_MATHLIB`).
- Scalar finite scores only.
- Sparse Vector / adaptive positive-release accounting not verified.

13 Open questions

Adaptive query order, limited positive releases, likelihood-ratio stability parameters, and full Sparse Vector theorems.

14 Connections to existing framework

Discovery IR → sealed CRP → System B → System 3 Lean (`Research.Operators.Threshold.Preservation`). Certificate: `lean/certificates/thresholding/threshold-output-preservation/`.

15 Verifier summary

Publication requires `LEAN_FULL` (sorry/admit = 0, `build_ok`). Prior computational-only paper archived under `_archive/v0-computational/`.

16 Appendix

Canonical statement (repository string):

> Let $x, x' \in \mathbb{R}$ and $\varepsilon \geq 0$ with $|x' - x| \leq \varepsilon$. For $A_T(x) = \mathbf{1}\{x \geq T\}$ with fixed $T \in \mathbb{R}$: (1) if $x \geq T + \varepsilon$ then $A_T(x') = 1$; (2) if $x < T - \varepsilon$ then $A_T(x') = 0$; (3) if $x \in [T - \varepsilon, T + \varepsilon)$ the output need not be invariant.

References

References

[1] Dwork, C., and Roth, A. *The Algorithmic Foundations of Differential Privacy.* Foundations and Trends in Theoretical Computer Science, 2014. (AboveThreshold / Sparse Vector as noisy thresholding primitives.)

[2] Repository implementation: `implementation/src/operators/thresholding/`.